Learning Trajectory of Two-Variable Linear Equations Using Problem-Based Learning Supported by Artificial Intelligence

ABSTRACT

Systems of linear equations in two variables remain a persistent challenge for students, particularly when modelling contextual situations into algebraic procedures. Many learners continue to rely on procedural routines, remain vulnerable to symbolic errors, and rarely engage in verification or reflective reasoning. This study aims to develop a learning trajectory for linear equations in two variables by integrating problem-based learning with the structured support of artificial intelligence to solve mathematical problems. The research employed a design research methodology, consisting of three phases: preliminary design, design experiment, and retrospective analysis, following a validation framework to refine the hypothetical learning trajectory. Participants were thirty-four eighth-grade students from a public junior high school in Indonesia. Data were collected through interviews, classroom observations, and documentation of student work. The findings indicate that students actively engaged in discussion, were more willing to ask questions, and interacted productively with the AI tool. Interviews confirmed that problem-based learning encouraged deeper reflection, while the AI provided immediate and personalised feedback. The validated trajectory demonstrates that problem-based learning supported by artificial intelligence can effectively enhance students' conceptual understanding and foster reflective reasoning in addressing non-routine mathematical problems.

Keywords: Artificial Intelligence, Educational Design Research, Educational Technology, Learning Trajectory, Problem-Based Learning.

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Abstract

Systems of linear equations in two variables remain a persistent challenge for students, particularly when modelling contextual situations into algebraic procedures. Many learners continue to rely on procedural routines, remain vulnerable to symbolic errors, and rarely engage in verification or reflective reasoning. This study aims to develop a learning trajectory for linear equations in two variables by integrating problem-based learning with the structured support of artificial intelligence to solve mathematical problems. The research employed a design research methodology, consisting of three phases: preliminary design, design experiment, and retrospective analysis, following a validation framework to refine the hypothetical learning trajectory. Participants were thirty-four eighth-grade students from a public junior high school in Indonesia. Data were collected through interviews, classroom observations, and documentation of student work. The findings indicate that students actively engaged in discussion, were more willing to ask questions, and interacted productively with the AI tool. Interviews confirmed that problem-based learning encouraged deeper reflection, while the AI provided immediate and personalised feedback. The validated trajectory demonstrates that problem-based learning supported by artificial intelligence can effectively enhance students' conceptual understanding and foster reflective reasoning in addressing non-routine mathematical problems.

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INTRODUCTION

A learning trajectory constitutes a practical, strategic pathway for helping students develop a deep, gradual, structured, and meaningful understanding of mathematical concepts, particularly in solving non-routine mathematical problems (Páez et al., 2022). The trajectory of problem-based learning is recognised as a pedagogical approach that fosters higher-order mathematical thinking, collaboration, and learner autonomy. Supported by artificial intelligence, it offers innovations in personalised learning that analyse students' difficulties with fundamental concepts of Linear Equations in Two Variables (LETV) and creatively adapt problem situations to address real-life issues (Sari et al., 2024; Widiatsih et al., 2020). The integration of PBL with AI has been shown to meaningfully strengthen students' computational and cognitive abilities. Findings from prior studies (Guo et al., 2025; Jalinus & Putra, 2024) indicate that AI-assisted problem-based learning significantly enhances students' problem-solving skills in mathematics. Further evidence (Feng et al., 2025; Kaswan et al., 2024) highlights that AI-based mathematics learning platforms effectively support personalised instruction by adapting content, modes of presentation, and levels of instructional scaffolding, demonstrating the considerable potential of AI to meet diverse student needs. Consistent with other studies (Han et al., 2024; Stankevich et al., 2025a), adaptive AI-supported mathematics learning also offers cost-efficient solutions, particularly in addressing competency gaps in developing countries. The integration of Photomath reflects a technology-driven pedagogical transformation in which AI functions not merely as a problem-solving tool but as a facilitator of mathematical thinking processes (Ahmad et al., 2025a; Vintere et al., 2024). This strategy aligns with the goals of 21st-century education, emphasising critical thinking, collaboration, and technological literacy. Further research is required to explore how such multi-AI integrations can be effectively implemented in classrooms, particularly to support the learning trajectories of LETV through problem-based learning.

Empirical evidence in the field shows that students' ability to solve non-routine mathematical problems remains low, especially in applying problem-solving strategies and providing justifications using alternative solution methods. While students may produce different solutions, they often find it challenging to present clear, logical reasoning to confirm their correctness (Kusuma & Untarti, 2020). Many students struggle to connect word problems with the right mathematical models. Instead of understanding the core concepts, they often memorise elimination or substitution procedures but fail to grasp the meaning of the symbols they use (Anjariyah et al., 2022; Herman & Fatimah, 2023). This condition reinforces the view that mathematics instruction should emphasise the interconnectedness between real-world contexts and symbolic representations in mathematical problem-solving (Tran et al., 2020). In classroom practice, the teaching of systems of linear equations in two variables (SLETV) often remains procedural and mechanistic, limiting students' opportunities to build a deeper conceptual understanding. Furthermore, issues of unequal access to technology persist, as students are required to purchase internet data packages, and the positive potential of digital tools is often overshadowed by teachers' concerns that students may rely on applications to obtain instant answers without understanding the underlying processes (Rong & Mononen, 2022). Ideally, the design of SLETV instruction should be structured as a systematic and meaningful learning trajectory supported by technology, allowing students to not only solve problems algorithmically but also link mathematical concepts to their everyday experiences.

The Problem-Based Learning (PBL) approach offers such opportunities by positioning students as active agents in the discovery and construction of prior knowledge. Research findings (Stankevich et al., 2025b; Vintere et al., 2024) indicate that Artificial Intelligence (AI) support in the learning process opens new avenues for personalising learning experiences, providing adaptive feedback, and dynamically strengthening students' understanding. Developing a problem-solving–oriented learning trajectory for systems of linear equations in two variables (SLETV) through technology integration is thus highly relevant. Advances in information and communication technology (ICT) have expanded the possibilities in mathematics education, with digital applications such as Photomath gaining popularity among students due to their ability to provide step-by-step solutions instantly (Ahmad et al., 2025b). According to (Perienen, 2020) the integration of ICT in mathematics instruction enhances effectiveness, engagement, and accessibility, while stimulating meta-strategic reflection. The literature on ICT in mathematics education underscores that technology use must be directed toward appropriate pedagogical functions. Recent research (Al-Barakat et al., 2025) highlights the importance of designing activities in which technology does not replace students' cognitive effort but instead supports their thinking processes. In the context of Photomath, the challenge lies in transforming its role from a "solution provider" to a "reflective partner" that helps students detect errors, compare strategies, and strengthen justifications. In other words, integrating Photomath requires precise ethical regulation to ensure that the technology truly supports meaningful learning.

Several challenges remain evident in the concept of systems of linear equations in two variables (SLETV). First, students often make mistakes in formulating equations from word problems, particularly in contextual tasks, where they struggle to correctly identify coefficients and constants (Utami & Kusumah, 2024). Second, students' justifications for algebraic steps are often weak, characterised by overly general reasoning that fails to reference the laws of equivalence (Adu-Gyamfi et al., 2025). Third, the phenomenon of strategy switching, in which students compare manual solutions with results generated by applications, has not been systematically examined (Ramazanova et al., 2021). Fourth, the ethical guidelines for using Photomath in classroom practice have yet to be clearly formulated, leading to resistance among teachers (Ahmad et al., 2025a). Finally, a validated learning trajectory for SLETV that explicitly includes Photomath in students' reflective processes remains missing (Ahmad et al., 2025b).

To address these gaps, this study seeks to develop a learning trajectory for systems of linear equations in two variables (SLETV) that combines context-based instruction, problem-based learning (PBL) strategies, and the ethical integration of Photomath. This research is significant because junior high school students still tend to rely on mechanical procedures, are prone to symbolic errors, and rarely engage in verification or reflective reasoning. By designing activities that require students to articulate justifications, detect errors, and compare manual work with application-generated solutions, instruction can be directed toward enhancing mathematical problem-solving abilities and reflective skills. The integration of a learning trajectory through problem-based learning supported by Photomath demonstrates both novelty and high relevance, as few studies have explicitly combined SLETV learning trajectories with AI-based PBL, positioning Photomath not as an instant solution provider but as a diagnostic and reflective tool. Thus, Photomath is framed as scaffolding that supports rather than replaces the thinking process. This design aligns with the findings of Nursyahidah et al. (2025) and Widiatsih et al. (2020), which suggest that technology should be treated as a didactic instrument that enriches mathematical interaction rather than as a mere substitute. This research aims to design a learning trajectory for SLETV concepts through problem-based learning supported by artificial intelligence. This study responds to teachers' practical needs while also contributing to the academic discourse on technology integration in mathematics education.

METHODS

Research Design

This study is grounded in design research in mathematics education, where the central aim is to develop and refine a local instruction theory through iterative cycles that tightly connect theory, designed tasks, and actual classroom activity. A hypothetical learning trajectory HLT is used as both a design tool and an analytic lens, consisting of learning goals, a sequence of tasks, and conjectures about students' reasoning and the teacher's moves. The work proceeds through preparation for the experiment, the teaching experiment, and retrospective analysis, with continual documentation of students' strategies, interactions, and emerging mathematical ideas to inform successive refinements of the trajectory (Gravemeijer & van Eerde, 2009).

The study adopts the validation studies approach, which comprises three phases: Preliminary Design, Design Experiment, and Retrospective Analysis. The goal is not only to produce a workable set of activities but to validate the conjectured trajectory by confronting it with classroom realities and systematically justifying design decisions with empirical traces from the class Plomp and Nieveen, 2013; Putri and Zulkardi, 2017. In our context, the HLT targets SPLDV learning with Photomath positioned as an ethically guided verifier. The technology is embedded at pre-specified checkpoints so that it supports reflection and diagnosis rather than replacing student thinking.

Preliminary Design

In this phase, we conduct a content and learner analysis of SPLDV in the target grade, review curricular demands, and map common difficulties in modeling from word problems to two linear equations. We then construct the HLT with four activities that move from contextual modeling to formal solution, structured verification, and interpretation. For each activity, we specify goals, tasks, expected informal and formal strategies, and teacher prompts. We also established an ethical use protocol for Photomath, which involves writing first, followed by checking, minimal assistance, required paraphrasing and rationale, and logging of every access. Design artifacts prepared include LKPD for four activities, observation guides, a comparison box for mine versus app steps, a short log of Photomath access, and rubrics for model quality, stepwise justification, and contextual argumentation. Expert review is used to scrutinize task clarity, cognitive demand, and the placement of technology checkpoints before piloting.

Design Experiment Pilot & Experiment

A small group pilot of eight to ten students is used to test the clarity of instructions, timing per step, and practicality of the Photomath protocol. We observe where students get stuck, whether prompts elicit the intended reasoning, and how often and why students invoke the app. Mismatches between the conjectured and actual strategies, for example, misreading coefficients or over-reliance on copying, are documented. Based on pilot evidence, we revise the wording of tasks, tighten the one-step-only rule for assistance, and adjust teacher prompts to foreground reasons for operations rather than procedures.

The revised HLT is enacted in a whole-class setting with a model teacher, while the researcher documents processes through field notes, video snippets as available, student artifacts, and app screenshots. Photomath is permitted only at predefined checkpoints to scan and verify written equations in Activity two, after a three-minute pause in Activity 3 with mandatory paraphrasing and justification, and at the end of Activity 4 for verification. Data captured includes the timing and frequency of app access per student, the feature used (scan, one step, or final check), how the external step was paraphrased, and what changes were made in the student's work afterward. This phase aims to surface the enacted learning trajectory ALT to be compared with the HLT.

RESULTS AND DISCUSSION

Preliminary Design

In the Preliminary Design phase, a preliminary study was conducted by interviewing a mathematics teacher who taught Grade VIII at the research site, as well as by observing and analyzing students' characteristics in terms of learning interests, learning styles, and readiness to engage with previously taught materials. The study also examined students' difficulties in understanding mathematical concepts and the challenges they faced during the learning process. From the teacher interview, it was reported that students experienced significant challenges in solving non-routine mathematical problems. Specifically, students struggled to interpret problem situations in word problems, apply problem-solving strategies, and carry out algorithmic solution processes. They also found it challenging to justify single-solution problems using multiple solution methods and to distinguish among various solutions through clear and logical reasoning. Moreover, students were not accustomed to solving non-routine mathematical problems, nor were they used to reading and understanding materials through worksheets and teaching modules before classroom sessions. Students were also unaccustomed to receiving worksheets tailored to their learning styles (kinesthetic, visual, and auditory) or to being given pre-learning assessments that could help them recognise the importance of preparation before face-to-face lessons and foster higher-order thinking skills. Likewise, formative assessments designed to promote reflective thinking, as well as post-learning assessments to measure understanding of mathematical concepts in real-life contexts, were rarely implemented. These challenges ultimately limited students' ability to develop meaningful understanding, leaving them to merely memorise formulas in ways that constrained effective thinking in mathematical problem-solving.

Based on the analysis of students' difficulties and challenges, the researcher designed a problem-based learning model supported by Photomath, utilising AI to assist in solving complex and confusing mathematical problems. This design serves as an alternative solution to address students' learning obstacles during instruction. Grounded in Piaget's constructivist theory and Vygotsky's sociocultural perspective, the approach was adapted for use with mathematics teachers at the school. One week before classroom instruction, students were provided with worksheets (LKPD) and teacher-developed learning modules tailored to their characteristics. The purpose of this preparation was to ensure that students had at least read, understood, diagnosed, and analysed the material and problems before meeting the teacher in class. As a result, students entered the classroom with questions and problem situations ready for discussion. This shifted the learning process into a problem-based sequence, beginning with students' own difficulties and inquiries, since they had already explored the material beforehand. In this way, students became actively engaged in constructing mathematical concepts to solve problems that connected to real-life experiences. Such engagement facilitated deeper conceptual understanding, made mathematics more accessible, and fostered meaningful learning grounded in authentic and contextual experiences.

The researcher collaborated with the mathematics teacher to design teaching modules, student worksheets, observation formats, problem-solving instruments, and Photomath-based interactive media, and to construct a Hypothetical Learning Trajectory (HLT) that connects the conceptual needs of Grade VIII SLETV with light yet evidence-rich classroom activities. The starting point was the initial finding that students frequently made errors in mapping contextual stories into mathematical equations, tended to be procedural in their problem-solving, and were not accustomed to verification or contextual interpretation. Accordingly, the HLT was designed into four sequential activities:

- A1: Understanding the problem context in word problems and conducting initial modelling (reading the context and constructing preliminary models).
- A2: Implementing problem-solving strategies by transforming sentences into two equations while choosing a plan with explicit justification.
- A3: Solving the algorithm step by step and verifying the process in a structured manner.
- A4: Demonstrating multiple solution methods, verifying outcomes, interpreting solutions, and reflecting on the ethical use of AI.

The role of Photomath was deliberately restricted to function as a verification/diagnostic partner rather than a source of answers: in A2 (to check equation formulation), in A3 only after ≥3 minutes of impasse (with mandatory paraphrasing and justification), and in A4 for final verification (with required screenshot and a comparison box titled *"my work vs. the application"*). The prepared instruments included worksheets with "transformation–justification" columns, Photomath access logs, comparison boxes, and reflection cards, all designed to make students' thinking processes visible. This design ensured that comparisons between the HLT and the Actual Learning Trajectory (ALT) in the retrospective analysis phase would be supported by clear traces of evidence. Table 1 summarises the activities, objectives, and conjectured student responses that form the core of the HLT. The *Objectives* column specifies the intended learning goals, while the *Conjectures* column presents expected responses alongside predicted errors and their corrective mechanisms.

Table 1. HLT Design for SLETV Supported by Photomath

Activity	Objectives	Conjectures
Activity 1 Understanding the problem context in word problems and initial modeling	• Understand the facts presented in the problem	• Some students understand accurately;• Some students make errors but can be corrected through unit-checking.
Activity 2 Implementing problem-solving strategies	• Transform two sentences from the problem situation into two distinct (non-redundant) equations; • Choose elimination, substitution, or graphical methods with one structural reason (e.g., coefficients are easily aligned, avoiding fractions).	• Majority of students succeed;• Some students make mistakes in planning problem-solving strategies they perceive as easier.
Activity 3 Step-by-step problem solving using Photomath support	• Carry out equivalent algorithmic steps with reasoning aligned to the problem situation	• Solution pathways become more organized; • Sign or distribution errors are quickly detected; • Some students switch strategies after comparing with the application.
Activity 4 Multiple-solution proof, verification, interpretation, and ethical reflection	• Use a graphical method to verify solution accuracy; • Interpret the meaning of ordered pairs (x, y) in context; • Document two solution methods using Photomath ethically.	• Some students struggle to construct alternative solution methods yielding the same answer; • Generally results are consistent; when discrepancies arise, errors are identified through the comparison box;Arguments become stronger; class rules established: “write first–check later, one step when stuck, mandatory paraphrase and justification.”

From Table 1, it can be seen that the outcome of this stage is a structured Hypothetical Learning Trajectory (HLT) consisting of four main activities. This stage is crucial to ensure that each step of instruction has clear objectives, predicted student responses, and is safe and ethical to use with Photomath. With the HLT in place, teachers and researchers gain an initial guide to anticipate possible student errors while also preparing instruments and artefacts that capture the learning process in detail. The Preliminary Design phase thus serves as a foundation for ensuring that subsequent classroom trials (teaching experiments) proceed in a focused and measurable way, generating rich data for analysis in the retrospective phase.

Design Experiment

The design experiment phase consisted of two stages: the pilot experiment and the teaching experiment. In the pilot experiment, the initial HLT design was tested on a small group of six students. The primary aim was to examine the clarity of instructions, time allocation, readability of the worksheets (LKPD), and students' adherence to the rules for using Photomath. Findings from this stage indicated several necessary revisions: the complexity of scientific language needed to be simplified for better comprehension, explicit unit checks had to be incorporated into the initial activity, and the use of Photomath had to be strictly limited to brief scan checks, one-step checks when stuck, and final verification. In addition, teacher instructions were revised to emphasize *reason-first prompts*—questions that encourage students to explain the rationale behind each step rather than merely performing procedures.

The next stage was the teaching experiment, in which the revised HLT was implemented with a whole class of 32 students. The teacher served as the model teacher, facilitating each activity in the planned sequence. The researcher served as observer, documenting the learning process, recording moments when students used Photomath, and collecting artefacts such as worksheets (LKPD), comparison boxes, usage logs, and reflection cards. At this stage, attention was focused not only on whether students arrived at the correct final answers, but also on how the Actual Learning Trajectory (ALT) compared with the predicted HLT. These data then formed the basis for retrospective analysis, aimed at assessing consistency, identifying surprises or deviations, and deciding on subsequent design revisions. The results of these revisions are presented in Table 2.

Table 2. Comparison Between Pilot Version and Revisions for the Teaching Experiment

Component	Pilot Version	Revision for Teaching Experiment
Task language	Sentences were relatively long and difficult to understand	Simplified into shorter and more concrete statements
Unit-check	Not explicitly included in Worksheet A1	Added a unit-check column so that students write variable units from the beginning
Photomath protocol	No clear time regulation	Clarified: scan-check for 60 seconds (A2), one step when stuck (A3), final verification with screenshot + comparison box (A4)
Teacher instructions	Teacher's questions tended to be procedural	Replaced with reason-first prompts such as "What principle of equivalence are you using?"

Component	Pilot Version	Revision for Teaching Experiment
Time allocation	Not standardized, some activities exceeded the time limit	More strictly regulated: A1 (12’), A2 (15’), A3 (20’), A4 (10’)

Based on the trial results and feedback from the mathematics teacher, the researcher and teacher agreed to revise Activity 2. This revision was necessary because, during the initial implementation, several challenges were identified, such as students misplacing constants or coefficients and struggling to understand the instructions for writing justifications of their chosen strategies. Consequently, the rules for using Photomath were clarified: it was limited to a brief scan check of no more than 60 seconds. Additionally, the instructions in the worksheets (LKPD) were simplified and supplemented with a short example to make them easier for students to follow. The revision also emphasised that students must provide structurally based justifications for their strategy choices, such as by aligning coefficients or avoiding fractions, so that the objectives of the activity could be achieved more effectively.

Teaching Experiment

The first activity was designed to guide students in understanding a contextual story problem and modelling it into a simple mathematical form. At this stage, students were asked to identify variables and their corresponding units, then formulate two statements of equality based on the information provided. The primary focus of this activity was to cultivate discipline in using units and to ensure that students could construct mathematical statements in natural language before transitioning to symbolic form. Student learning activities for this stage are presented in **Figure 1**.

Figure 1. The First Activity (illustration of classroom setting)

The illustration in **Figure 1** depicts the classroom atmosphere when the teacher introduced the first activity to the students. The teacher began by presenting a simple context involving the purchase of two product packages at the school canteen, then guided students in identifying the key information from the problem. Students proceeded to read and work on the worksheet (LKPD), writing down the variables with their units and formulating equality statements based on the given information. See Table 3

Table 3. Problem Statement from the Task

Package	Composition	Total	Statement in Sentence Form
Package A	2 bottles of juice A + 3 bottles of juice B	Rp150,000	*"The total amount paid = 2 × price of juice A + 3 × price of juice B."*
Package B	1 bottle of juice A + 2 bottles of juice B	Rp80,000	*"The total amount paid = 1 × price of juice A + 2 × price of juice B."*

The problem presented to students was a simple school canteen context. Students were asked to determine the prices of two types of juice using information from two different packages. From this context, students were guided to define variables, specify units, and write equality statements before moving to symbolic form.

Teacher: "Who can identify the variables from this story?"
Student: "x is the price of one bottle of juice A, in rupiah."
Teacher: "Good. What about the second variable?"
Student: "Y is the price of one bottle of juice B, also in rupiah per bottle."
Teacher: "Now, try to state the equality for Package A."
Student: "The total amount paid is equal to two times the price of juice A plus three times the price of juice B."

The brief interview above illustrates that students were able to grasp the core of the context by assigning appropriate variables and units, and then expressing the relationship in natural language. This indicates that the process and teacher guidance were effective in preventing early mistakes, such as misinterpreting variables or neglecting units. The teacher–student interaction also underscores the importance of the first activity as a foundation: without an accurate model at this stage, students would encounter difficulties when transitioning to symbolic equations in subsequent activities.

Activity 2. Sentences to Two Equations & Strategy Choice

The second activity serves as a pivotal point in the learning trajectory, as it is the first time students engage with a contextual problem, transitioning from verbal statements to formal equations. Example word problem;

At the school canteen, there are two packaged options of small bottles of juice: starfruit and apple. Rudi buys two bottles of apple juice and three bottles of starfruit juice for Rp150,000. Meanwhile, Toni buys one bottle of apple juice and two bottles of starfruit juice, paying Rp80,000. Determine the price of one bottle of apple juice and one bottle of starfruit juice. Solve this problem using two different methods, ensuring that the reasoning in both is correct and logical.

In this activity, students were asked to transform the two equality statements they had previously constructed into symbolic form as a pair of linear equations. The collected worksheets revealed varying levels of quality: while most students were able to write the equations correctly, some still confused constants with coefficients. This finding aligns with the initial conjecture that the transition from verbal statements to symbolic form represents a critical stage that often generates difficulties. To address this, the teacher instructed students not only to write the equations but also to complete a short table of values to test the consistency of the relationship between x and y. This step was essential because it helped students connect the symbolic representation to the real-world context. On the worksheets, some groups successfully filled in the table, while others discovered inconsistencies when their calculated totals did not match the stated package prices. Students' answers are summarised in Table 4.

Table 4. Example of Student Responses

Representation	Student Response
Equation 1	2x+3y=150,000
Equation 2	x+2y=80,000
Value Table (example)	(x=60,000, y=10,000); (x=50,000, y=15,000)
Strategy Choice	Elimination, because the coefficients of x are easy to align

In the next stage, students were asked to choose a solution strategy: elimination or substitution. Analysis of the worksheets revealed that students' reasoning behind their strategy selection began to emerge more clearly. Some students wrote, *"elimination is faster because the coefficients of x are easy to align,"* while others stated, *"substitution is chosen to avoid fractions."* This phenomenon is an early sign of evaluative thinking, in which students begin to make structural considerations rather than relying solely on procedural habits. However, not all students initially wrote the equations correctly. On several worksheets, misplacements of coefficients or constants were observed, for example: 2x+y=80,000, 2x+y=80,000. The teacher then provided a simple guiding question: *"How many bottles of A and B are in this package?"* This contextual prompt encouraged students to re-examine their work, and many subsequently corrected it to x+2y=80,000x+2y=80,000. This demonstrates how creativity unfolds: common errors can be reduced through focused yet straightforward guiding questions.

Interestingly, in several groups, students engaged in rather intense discussions before deciding on a strategy. Some initially chose substitution, but after comparing the number of steps in the table, they switched to elimination. This phenomenon of method switching reflects the emergence of creativity: students began to prioritise efficiency over simply solving the problem in any way. Observation notes indicate that the teacher allowed these small debates to continue so that students could practice justifying their choices. Students thus learned through the experience of making mistakes in class and correcting them with the appropriate solutions. From the researcher's perspective, the classroom atmosphere during this activity was lively. The discussions were noticeably more animated than in Activity 1, as students felt challenged to choose a strategy. Some provided simple justifications such as "easier" or "faster." Still, the teacher emphasised that reasons must be linked to the structure of the equations, for example, aligned coefficients or the avoidance of fractions. In this way, the activity functioned as a trigger for fostering more reflective mathematical thinking.

Artefact analysis revealed a general pattern: students who wrote the equations correctly from the outset found it easier to provide structural reasons for their chosen strategy, whereas those who initially misplaced coefficients tended to be more cautious in selecting a method after correcting their errors. This distinction shows that mistakes are not obstacles but starting points for encouraging reflection. The revisions documented on the worksheets provided tangible evidence that students were learning from diagnosed errors. See Table 5.

Table 5. Comparison of HLT and ALT in Activity 2

Aspect	HLT (Prediction)	ALT (Realization)
Equation formalization	Students write two correct linear equations (SLETV) from the contextual sentences	Majority correct; some errors in coefficients/constants, later corrected through teacher guidance
Strategy justification (analysis–evaluation)	Students choose elimination/substitution with a brief structurally based reason	Reasons appeared such as *"align the coefficients"* or *"avoid fractions"*
Coefficient errors (analysis–creation)	Potential misplacement of numbers can be minimized with teacher guidance	Actual errors (e.g., 2x+y=80,000) were corrected after contextual questioning
Method change	Students remain with their initial strategy	Switching of methods occurred after group discussion or comparison with solution steps

Activity 2 demonstrates that the transition from context to equations is not merely a symbolic exercise but a formative moment in developing mathematical thinking. The conjectures of analysis, evaluation, and creativity are evident: students articulated reasons for their chosen strategies, common errors were reduced through teacher scaffolding, and method switching occurred following peer discussions. The figures presented from students' worksheets illustrate these dynamics, highlighting that this activity not only stimulated interest in learning but also trained students to confront difficulties reflectively.

Activity 3. Stepwise Solution (Manual → Bantuan Terbatas → Verifikasi)

Activity 3 represents the core of the learning trajectory for systems of linear equations in two variables (SLETV), as this is the stage where students are truly challenged to solve the system of equations they have constructed. It was not sufficient for students to merely state the model; they were required to detail every step of the solution systematically, providing clear justifications in the *"Transformation"* and *"Reason"* columns of the worksheet (LKPD).

In the initial phase, most students started well, such as rewriting the two equations and selecting elimination by aligning the coefficients of variable *x*. However, as the steps progressed, several groups began to show confusion, especially when maintaining consistent signs or substituting results back into the original equations. This revealed that the transition from model to formal solution is a critical point requiring precision, making the teacher's role in observation and timely guidance essential.

Observations showed that students' confusion was not only technical but also conceptual. For instance, some students performed elimination correctly but could not explain why the step was algebraically valid. Others provided overly general reasons such as *"to make it easier"* or *"so it is faster,"* reflecting that they had not internalised the principle of equivalence in equations. On the other hand, some groups consistently offered sound justifications, such as writing *"multiplying both sides by the same number does not change the equality."* This variation is noteworthy, as it illustrates how the activity encouraged heterogeneous thinking: some students had begun to reason reflectively, while others remained procedural.

Procedural errors were fairly common, particularly sign errors in subtraction or mistakes in distributing numbers. For example, one student wrote 2x+4y–(2x+3y)=70,0002x+4y–(2x+3y)=70,000, which should have simplified to y=10,000y=10,000. Such mistakes became apparent when the *"Reason"* column did not align with the numerical result—e.g., the written reason *"subtracting the two equations"* was inconsistent with the outcome. The teacher responded by encouraging students to retrace their steps, while some opted to use Photomath after three minutes of impasse. In line with the rules, they were only permitted to view one subsequent step. Students then rewrote their corrections with paraphrased notes, such as *"it should be $4y-3y=y4y-3y=y$, not $y=70{,}000y=70{,}000$."* This mechanism positioned the application as a diagnostic instrument rather than a substitute for thinking.

Another interesting phenomenon was method switching. Some students initially chose substitution because they felt more comfortable, but after observing the shorter elimination pathway suggested by Photomath, they shifted methods. This switch did not occur automatically; it emerged through group discussion. For example, one student argued that substitution produced complex fractions, while another demonstrated that elimination reached the solution more quickly. Eventually, the group agreed to change strategies. This confirms that controlled integration of Photomath can stimulate strategic reflection rather than mere answer copying. Thus, the conjecture regarding possible method switching was clearly validated in the classroom. See **Table 6** for further details.

Table 6. Comparison of HLT and ALT in Activity 3

Aspect	HLT (Prediction)	ALT (Realization)
Manual steps & justification (creation)	Students write equivalent transformations with justifications at each step	Most students provided justifications, but some remained general (e.g., *"to make it easier"*), without referencing the law of equivalence
Procedural errors	Potential sign/distribution errors may occur and can be corrected through verification	Actual sign and distribution errors occurred; quickly diagnosed through the "reason" column and one-step use of Photomath
Method change	Students remain with their initial strategy (elimination/substitution)	Strategy switching occurred after discussion and comparison with application steps

Compared with the HLT predictions, the ALT data reveal both consistency and surprises. Overall, the conjecture on analysis was confirmed: students did provide reasons in their worksheets, though the quality varied. The conjecture on evaluation was also supported, as sign and distribution errors occurred frequently, yet these became critical moments where students learned to self-correct with minimal guidance. Meanwhile, the conjecture on creativity exceeded expectations, since strategy switching occurred more often than anticipated and even became a source of productive group discussion. This demonstrates that HLT predictions are not rigid boundaries but rather starting points for understanding classroom dynamics.

This activity highlighted the importance of making thinking processes transparent. The "Transformation Reason" column proved to be more than just an assessment tool; it functioned as a mirror that revealed the quality of students' reasoning. Through this column, teachers could immediately detect inconsistencies between procedures and justifications, then provide timely prompts. Without such an instrument, sign errors might only be discovered at the end when results failed to match, whereas with it, errors were detected earlier and corrected promptly. This underscores the academic contribution of designing instruments that focus on reasoning rather than solely on final answers.

The classroom atmosphere further confirmed the effectiveness of Photomath regulations. No student opened the application immediately; they first attempted the task independently and waited at least 3 minutes before consulting it when they were stuck. Thus, the application functioned as intended: a reflective scaffold to help identify errors or suggest alternative methods. This supports the argument that technology can be integrated ethically and in a controlled manner, provided that explicit rules are consistently enforced. Such findings offer practical reassurance for teachers concerned that applications like Photomath may reduce students' willingness to think.

Activity 3 thus serves as evidence that solving SLETV can be a platform for cultivating procedural accuracy, justification skills, and strategic reflection. The three main conjectures, analysis, evaluation, and creativity, were vividly observed in classroom practice: reasoning quality was diverse, procedural errors became opportunities for reflection, and method switching fostered productive discussion. From an academic perspective, these results affirm that design research can capture fundamental classroom dynamics while offering a model for integrating Photomath that does not undermine student autonomy but rather strengthens it.

Activity 4. Verification, Contextual Interpretation, and Ethical Reflection

The final activity was designed to bring SLETV's solutions back into the context of the story while simultaneously teaching students how to use Photomath ethically. After obtaining the solution (x,y)(x,y), students were asked to substitute it back into the two original equations to verify correctness, then express the meaning of the solution in natural language, for example: *"the price of juice A is Rp60,000, and the price of juice B is Rp10,000."* At this stage, Photomath was permitted for final verification under strict conditions: students had to take a screenshot of the application's result, then complete a comparison box with two columns, *"my answer"* and *"application's answer"*, followed by a brief reflection line stating *"what is the same/different, what I learned."* In this way, technology was positioned as a partner for reflection rather than a substitute for cognitive effort.

Figure 3. Student Work vs. Photomath Output

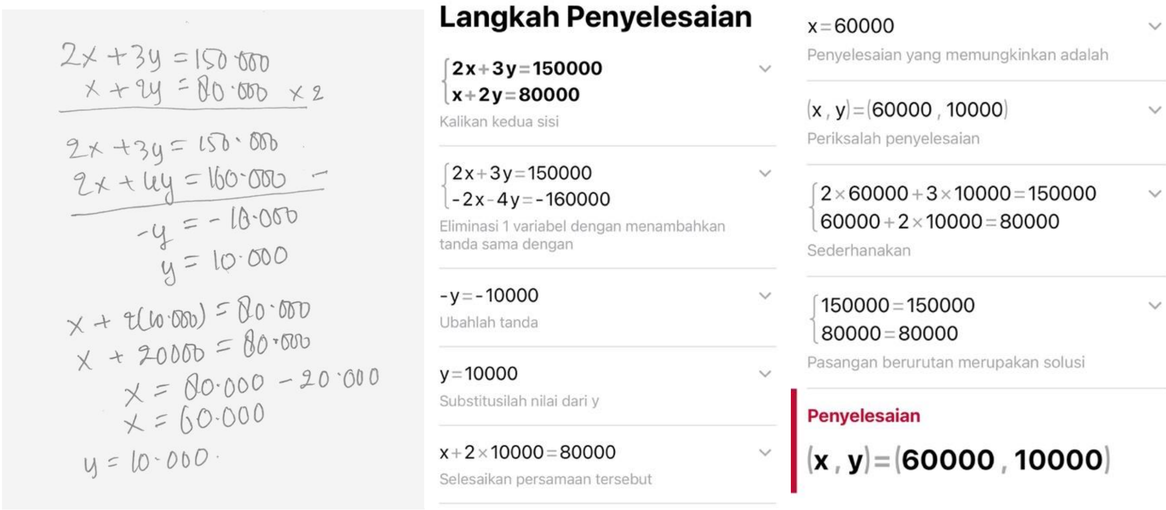

Figure 3 presents a comparison box illustrating the different solution paths taken by students and the application. Although the final results were identical (x=60,000, y=10,000), the methods differed: students employed elimination, while the application used substitution. From the researcher's perspective, this figure underscores that technology can stimulate strategic awareness: students recognised that there are multiple solution methods and were encouraged to weigh the efficiency of each approach rather than merely pursuing the correct answer. Thus, the comparison box served a dual function: both as a tool for verifying results and as an instrument for reflecting on the process.

Student quotations enrich the interpretation of this figure. For instance, the statement *"I made a sign error in the last step, but after checking with the application, I realised where the mistake was"* shows that the application acted as a procedural error detector, enabling students not only to receive passive correction but also to identify the exact point of error. The remark *"I used elimination, the application used substitution; it turns out there is more than one solution method"* indicates the emergence of strategic awareness that SLETV can be solved in multiple ways, and that method choice should be based on step efficiency. Meanwhile, the reflection *"My rule: check with Photomath only at the end, not at the beginning"* illustrates the development of self-regulation and ethical use of technology.

This analysis demonstrates that through verification activities, students not only validated mathematical results but also cultivated metacognition and a reflective stance toward technology. These findings, drawn from both comparison box artefacts and student quotations, provide a strong foundation for comparing the predicted HLT with the actual ALT. A summary of this comparison is presented in Table 7, highlighting how verification, contextual interpretation, strategic reflection, and rules for Photomath use were realised in classroom practice. See Table 7.

Table 7. Comparison of HLT (Prediction) and ALT (Findings) in Activity 4

Aspect	HLT (Prediction)	ALT (Findings)
Solution verification	Students substitute back into the original equations to confirm results	Most did so correctly; sign errors were corrected with the help of the comparison box
Context interpretation	Students write the meaning of the solution pair in narrative form	Nearly all expressed that "the prices make sense" and re-checked with the original equations
Strategy reflection	Not explicitly predicted	The comparison box triggered discussions on the efficiency of elimination vs. substitution
Photomath rules	Students formulate two personal rules for application use	Almost all wrote rules such as *"write first–check later," "one step only,"* and *"reasons required"*

The enactment results show that the HLT was validated primarily, particularly for numerical verification and contextual interpretation. However, the ALT added a reflective element that was not fully anticipated when comparing strategy efficiency using the comparison box. The fact that students were able to articulate personal rules for using Photomath also emerged as an important finding: the application functioned not only as a verifier but also as a medium for fostering ethical and metacognitive awareness. These findings expand the contribution to local instruction theory by demonstrating how SLETV can be taught with technology positioned not as the primary problem-solver, but as a reflective partner that helps students think more critically and responsibly.

The Research Findings

The research findings from the design experiment phase resulted in a learning trajectory that underwent two experimental stages: a pilot experiment and a teaching experiment. The learning trajectory for teaching the concept of SPLDV using Photomath, developed in this study, is presented in Table 8.

Table 8. Learning Trajectory for the Concept of SPLDV with Photomath

Activity Stage	Learning Target	Students can…
Understanding the problem context	Read the SLETV word problem, define variables and units	Write variables x,y with their units and construct two equality statements from the problem text
Formulating equations & choosing strategy	Translate sentences into two linear equations and select a strategy (elimination/substitution)	Write two correct SLETV equations and provide a simple justification for the chosen strategy
Stepwise solution with justification	Solve SLETV step by step with justification for each transformation	Explain transformations, detect sign/distribution errors, and correct them with limited Photomath support

Activity Stage	Learning Target	Students can…
Verification & reflection	Verify results, interpret the context, and formulate ethical rules for Photomath	Connect solutions to the context (e.g., reasonable prices), compare methods, and create personal rules for application use

The next step was the retrospective analysis, which compared the HLT with the actual learning process from the teaching experiment. Overall, there were no significant changes in the core activities designed to understand the concept of SPLDV. However, there were unique and noteworthy findings in students' responses to the learning process. These findings support, complement, and serve as references for comparison between HLT and the actual learning process. Table 9 presents these new findings for each learning activity on the SPLDV concept using PBL with the assistance of Photomath.

Table 9. Findings in Each Activity

Activity	Finding
Activity 1	Students demonstrated interest in contextual tasks and were able to define variables with units, although some confused variable meaning at first. Teacher's emphasis on unit-check reduced this error.
Activity 2	Students formulated two equations correctly in most cases; however, some misplaced constants/coefficients. Peer-check and cue questions helped correction. Students began to articulate reasons for choosing elimination or substitution.
Activity 3	Students wrote stepwise transformations, but quality of justifications varied. Errors in signs and distribution were frequent but quickly diagnosed. Some groups switched from substitution to elimination after reflection on efficiency.
Activity 4	Students successfully verified solutions and interpreted context. Comparison boxes revealed differences in methods (elimination vs substitution). Most wrote ethical rules for Photomath use, showing reflective and metacognitive growth.

The findings of this study indicate that students' initial difficulties in formulating mathematical models from contextual problems can gradually be addressed through carefully designed activities (Kristiani, 2024). Students' struggles in defining variables and units point to a gap in their ability to connect textual information with symbolic representations (Mardiah et al., 2021). By embedding the unit-check procedure into the Hypothetical Learning Trajectory (HLT), errors are not treated as failures but as opportunities to deepen understanding. This aligns with the principles of Educational Design Research (Mardiah et al., 2021; Sari et al., 2024; Utari & Putri, 2024) which emphasize that interactive design not only solves practical classroom problems but also contributes theoretically by documenting the transformation of interventions in real practice.

The process of model formulation in Activity 1 highlights the importance of representation and connection as integral components of mathematical competence (Jingga, 2019). Students' progress from ambiguous expressions to well-defined variables reflects not merely procedural skills but also conceptual reinforcement regarding how symbols represent quantities in context. This subtle shift underscores the dual role of design-based research: to provide practical support for students while enriching theoretical insights into the conditions that enable meaningful mathematical representation (Dannels, 2018). Thus, the seemingly simple activity of defining variables became a significant entry point in developing students' representational fluency.

Activity 2 highlights that the long-standing issue of weak algebraic justification can be addressed through intentional instructional design. Rather than allowing justification to remain a purely mechanical explanation, students were encouraged to articulate their reasoning in writing, supported by peer dialogue and teacher prompts. While some still provided general reasons such as *"to make it easier,"* others began to reference structural properties, for example, aligning coefficients or avoiding fractions. This indicates a shift from procedural dependence toward relational understanding, consistent with the international discourse that positions reasoning and communication as core components of mathematical proficiency (Syarifuddin & Atweh, 2022; Tshering, 2024; Veith et al., 2022). Thus, this design-based intervention established a space where justification itself became an object of learning, rather than just a byproduct of problem solving.

The emergence of peer correction further enriched this picture. When one student wrote the equation $2x+y=80,0002x+y=80,000$, another quickly corrected it by pointing to the context: *"There are two bottles of B, so it should be 2y."* This situation illustrates that justification, when situated in a collaborative environment, is strengthened by peer accountability and contextual reasoning. These findings are consistent with Mustaffa et al. (2016), who emphasize the teacher's role as a facilitator in creating space for mathematical argumentation. From an Educational Design Research (EDR) perspective, this confirms that the conjectured support mechanism of peer-checking can indeed foster mathematical justification practices in authentic classroom settings.

Activity 3 highlights a phenomenon that is relatively underexplored in the literature, strategy switching, which occurs when students identify discrepancies between their manual work and digital outputs. While the HLT predicted verification activities, classroom observations revealed that students shifted from substitution to elimination based on considerations of efficiency. This unanticipated phenomenon proved valuable, as it demonstrated how technology can stimulate strategic reflection. This aligns with (Renner et al., 2020) who observed that ICT tools, when used judiciously, can trigger meta-strategic reflection. In this context, Photomath functioned not merely as an error detector but also as a catalyst that encouraged students to evaluate their strategic choices.

The phenomenon of strategy switching carries significant implications for mathematics education theory. It highlights the pragmatic dimension of mathematical competence, where problem solving is not merely about arriving at a solution but also about selecting an efficient and appropriate method. Students' debates on whether elimination or substitution was shorter illustrate the development of reflective problem-solving skills, which (Kablan & Günen, 2021; Kholid et al., 2022; Syamsuddin et al., 2020) classify as higher-order competencies. Documenting such moments shows how design-based interventions can trigger metacognitive engagement while also providing empirical evidence that ICT, when governed by strict rules, enriches rather than diminishes students' mathematical thinking.

Activity 4 directly addressed concerns regarding the absence of ethical guidelines for technology use in mathematics learning. The use of the comparison box required students not only to compare manual solutions with those generated by the application but also to formulate personal rules for using Photomath. Interestingly, most students articulated principles such as *"write first, then check," "use one step only when stuck,"* and *"always provide reasons."* These self-generated rules demonstrate the emergence of metacognitive awareness and digital ethics, dimensions rarely reported in prior studies. This aligns with the discourse in ICT in Mathematics Education (Saritepeci & Yildiz Durak, 2024; Sosa-Alonso et al., 2025; Toker & Akbay, 2022), which emphasizes that technology integration must be accompanied by the cultivation of critical and reflective dispositions.

Student reflections further reinforce this analysis. Statements such as *"I made a sign error in the final step, but after checking with the application, I realized where the mistake was"* show how procedural errors became learning opportunities through digital verification. Meanwhile, *"I used elimination, the application used substitution; it turns out there is more than one way,"* reveals a new awareness of strategy diversity. From a design perspective, this validates the hypothesis that constrained use of technology can foster metacognitive awareness. From the standpoint of mathematical competence, it illustrates the integration of reasoning, communication, representation, and problem solving—the very dimensions emphasized by (Ate et al., 2024; Filiz & Gür, 2025; Sosa-Alonso et al., 2025) as the core of mathematical proficiency.

The validated learning trajectory, as outlined in Table 8, represents not merely a sequence of activities but a theoretical model for integrating ICT into algebra instruction. Each stage—from contextual modeling, equation formulation, and step-by-step solution, to verification and reflection—provides an empirical foundation for a local instruction theory of teaching SLETV with technological support. This aligns with the aims of Educational Design Research (Ate et al., 2024; Damrongpanit et al., 2023; Junhong et al., 2025; Sadik, 2021; Saritepeci & Yildiz Durak, 2024) which seeks to address practical classroom challenges while advancing our understanding of how technological support can be integrated ethically and pedagogically. Validation of the trajectory through two cycles of experimentation ensured that the model is not merely hypothetical but firmly grounded in classroom reality. The findings offer rich situational evidence that informs both practice and theory. Errors in equation formulation were addressed through the unit-check mechanism; weaknesses in justification became opportunities for learning argumentation; strategy switching was identified as a reflective potential; ethical rules for Photomath use emerged from student reflection; and the trajectory itself was validated through iterative experimentation. Collectively, these findings contribute to international conversations in mathematics education on the roles of EDR, mathematical ability, and ICT integration.

This study extends the development of local instruction theory (LIT) by demonstrating how ICT integration can be pedagogically and ethically managed in algebra instruction. The validated trajectory provides evidence that technology can trigger significant phenomena—such as structural justification and strategy switching, that have previously received limited attention in the literature. These findings affirm the view that mathematical ability encompasses not only procedural skills but also reasoning, representation, connection, and communication, as well as practical contributions. The study offers concrete guidance for teachers on integrating digital applications into the classroom. Clear usage protocols—"write first, check later," "only one step of help when stuck," and "reasons must always be written"—proved effective in fostering learning discipline while building students' ethical awareness of technology. Teachers gain step-by-step examples of activities ready for classroom implementation, while students gain learning experiences that emphasize not only outcomes but also reflective processes. This study focused exclusively on the integration of technology in the context of SLETV. Further research is needed to explore its application across other mathematical topics. In addition, school facilities and students' readiness for technology use remain factors that must be carefully considered.

CONCLUSION

The design and validation of a learning trajectory for the topic of Systems of Linear Equations in Two Variables (SLETV) at the junior high school level was carried out using a problem-based learning approach, ethically integrated with digital technology. Through the framework of Educational Design Research (EDR), this study emphasizes that teaching SLETV should not be limited to algebraic procedures but must also involve contextual problem solving, justification of steps, and reflection on the use of technology. The findings indicate that the designed trajectory, which consists of contextual modeling, equation formulation, step-by-step solution, and verification and interpretation, can support students in reducing modeling errors, strengthening structural reasoning, and detecting procedural mistakes. The presence of a digital verification application, when regulated at specific points, proved effective as a reflective scaffold rather than a mere answer provider. Thus, the use of technology in SLETV learning can be constructively directed to support students' thinking processes. Further research is recommended to extend the trajectory model to other mathematical topics, explore a wider variety of digital applications, and test its effectiveness across more diverse classroom contexts. In this way, the contribution of this study goes beyond SLETV and provides direction for broader ICT integration in mathematics education.

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